Vertiport Planning in Chicago

Mathematical model, working document v1.1

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Abstract

This document describes the optimization model of the project. The model builds on Wang, Jacquillat and Vaze (2022), "Vertiport Planning for Urban Aerial Mobility: An Adaptive Discretization Approach," *M&SOM* 24(6):3215–3235. Their framework combines facility location with demand that depends on travel times. It also uses a distributionally robust demand function. Below, "the paper" refers to this article. This project uses a simplified version without the operations layer. It asks two questions.

RQ1: What changes when customer expectations are estimated for each origin–destination pair from observed rideshare fares and times, instead of being the same for all?

RQ2: How robust is the network to demand at different times of day? What does a network planned for a single period lose over a year?

1 Overview

The model has two layers. The *precomputation* layer (Python) converts trip records into travel times and switching shares for every possible routing. All nonlinear relations are computed there. The *optimization* layer is a mixed-integer linear program solved with Gurobi. It chooses the vertiport sites and routes each origin–destination flow. All reported runs maximize the number of air-taxi passengers. The profit version uses the margins of Section 4.5 as weights. Figure 1 shows the pipeline.

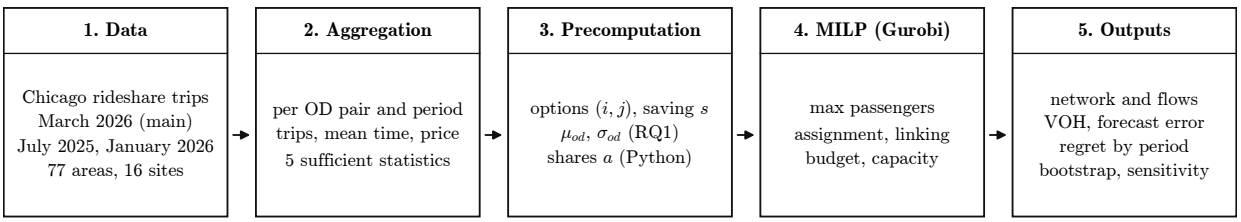


Figure 1: From trip records to the network and the RQ1/RQ2 metrics. t: ride time; q: ride price; s: relative time saving; a: switching share.

Figure 2 shows the choice for one origin–destination (OD) pair in one period. Each traveler either stays on the ground (rideshare) or flies through an open vertiport pair (i, j).

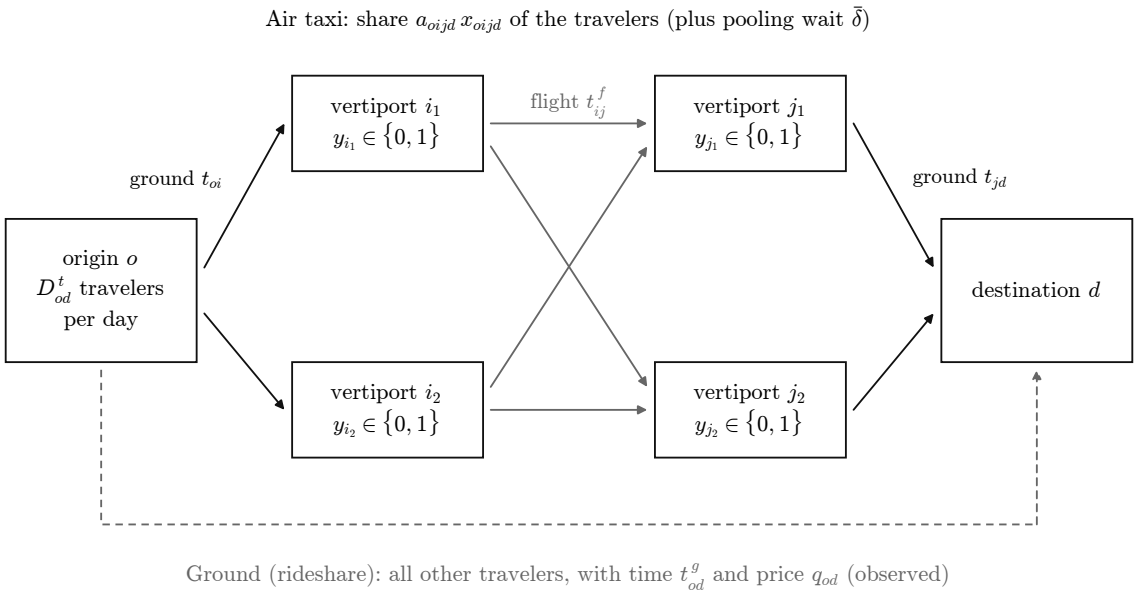


Figure 2: Routing options for OD pair (o, d) in period t. Black arrows are ground legs to and from vertiports. Grey arrows are flight legs. The dashed arrow shows the travelers who stay on the ground.

2 Sets and indices

Symbol	Index	Meaning
$\mathcal{Z}$	o, d	The 77 Chicago community areas (origins and destinations).
$\mathcal{V} \subseteq \mathcal{Z}$	i, j	Candidate sites. Main runs: 16 (O'Hare, Midway and Vertiport Chicago at exact locations, five high-demand areas, eight areas with cluster centers of Volakakis and Mahmassani 2024). Robustness runs: all 77 areas.
$\mathcal{T}$	t	Periods $\{\text{am}, \text{mid}, \text{pm}, \text{oth}, \text{we}\}$: weekday 7–10, 10–16, 16–19, other hours; weekend.
$\mathcal{P} \subseteq \mathcal{Z} \times \mathcal{Z}$	(o, d)	OD pairs kept per period: $o \neq d$, mean ground time ≥ 20 min, ≥ 1 trip per day.
$\mathcal{K}_{od} \subseteq \mathcal{V} \times \mathcal{V}$	(i, j)	Useful vertiport pairs for (o, d) (Section 4.2).

3 Data and parameters

3.1 Observed in the Chicago rideshare data

Source: City of Chicago, Transportation Network Providers trips (Socrata `6dvr-xwnh`). The main results use March 2026. The seasonal check uses July 2025 and January 2026. Trips are aggregated by pickup area, drop-off area and period, one day at a time. Shared trips, trips under 60 seconds or over 3 hours, and trips without a fare or an area are removed. Ground time is `trip_seconds`. Timestamps are rounded to 15 minutes. Ground price is `fare` + `additional_charges`. Fares are rounded to \$2.50.

Symbol	Meaning	Unit
D^t_{od}	Rideshare trips $o \to d$ per day of period type t	trips/day
$t^{g,t}_{od}$	Mean ground time $o \to d$ in period t	min
q_k, t_k	Price and ground time of trip k	\$, min
$m^{(1..5),t}_{od}$	Sufficient statistics $\mathbb{E}[1/t]$, $\mathbb{E}[q/t]$, $\mathbb{E}[1/t^2]$, $\mathbb{E}[q/t^2]$, $\mathbb{E}[q^2/t^2]$ (Section 5)	various
ℓ_{od}, ℓ_{ij}	Straight-line (haversine) distance between centroids or sites	miles

3.2 Assumed (with sensitivity ranges)

Values come from public sources.

Symbol	Meaning	Default	Range
v	eVTOL cruise speed (240 km/h for NYC)	200 km/h	150–240
t_{oh}	Fixed time per flight (boarding, takeoff, landing)	10 min	6–20
$\bar{\delta}$	Pooling wait (endogenous in the paper)	5 min	2–10
t_0	Ground leg within the vertiport's own zone	5 min	3–8
h	Transfer ("hassle") penalty added to air-taxi time	0 min	0–15
$s_{\min}$	Minimum relative time saving for an option	0.15	0–0.15
$t_{\max}$	Maximum ground leg to or from a vertiport	25 min	15–30
k	Nearest vertiports considered at each trip end	3	2–5
μ, σ	Uniform expectations (NYC only; Chicago: Section 5)	0.40, 0.20	μ: 0.33–0.40
P_0, r, b_0	Base fare, rate per mile, median value of time	see Table 1	–
CV_b	Coefficient of variation of the value of time	0.84	–
κ, c^g, g_i	Flight cost per mile, ground-leg cost, daily site cost (profit version)	not used	–
w_t	Days per year of period type t	250 (weekday), 115 (weekend)	–

Scenario	P_0	r (/mile)	b_0 (/hour)	Basis
Launch	\$15	\$6.00	\$62.00	near-term fare estimates; airport and business travelers
Launch, pessimistic	\$15	\$12.00	\$62.00	current helicopter shuttle prices
Adopter (main)	\$15	\$3.50	\$31.00	operator targets; air-taxi survey of Boddupalli et al. (2024)
Mature	\$15	\$2.50	\$20.10	long-run fare estimates; USDOT personal local travel

Table 1: Fare and value-of-time scenarios (2024 dollars). The rate per mile applies to the straight-line OD distance ℓ_{od}. The value of time is lognormal, with median b_0 and $\mathrm{CV}_b = 0.84$.

4 Precomputation

4.1 Ground and flight times

Ground legs. For zones u and v in period t:

$$t^t_{uv} = \begin{cases} t_0 & \text{if } u = v, \\ \text{observed mean } t^{g,t}_{uv} & \text{if } \geq 2 \text{ trips per day}, \\ \alpha^t + \beta^t \, \ell_{uv} & \text{otherwise (linear fit on pairs with } \geq 5 \text{ trips per day).} \end{cases} \tag{1}$$

Flight legs. With ℓ_{ij} in km:

$$t^f_{ij} = \frac{60 \, \ell_{ij}}{v} + t_{\mathrm{oh}}. \tag{2}$$

4.2 Useful vertiport pairs

Let $N^{\mathrm{acc},t}(o)$ be the k sites closest to o by ground time, with $t^t_{oi} \leq t_{\max}$. Let $N^{\mathrm{egr},t}(d)$ be the k sites closest to d, with $t^t_{jd} \leq t_{\max}$. Then

$$\mathcal{K}^t_{od} = \big\{(i,j) \in N^{\mathrm{acc},t}(o) \times N^{\mathrm{egr},t}(d) \ : \ i \neq j\big\}. \tag{3}$$

4.3 Air-taxi trip time and time saving

The trip time follows equation (5) of the paper, but with a fixed pooling wait:

$$\tau^t_{oijd} = t^t_{oi} + \bar{\delta} + t^f_{ij} + t^t_{jd} + h, \tag{4}$$

$$s^t_{oijd} = 1 - \frac{\tau^t_{oijd}}{t^{g,t}_{od}} \qquad \text{(relative time saving).} \tag{5}$$

4.4 Demand: the share of travelers who switch

As in Section 4.4 of the paper, a traveler on (o,d) compares two utilities: $V - P_{od} - b\,\tau$ for the air taxi and $V - q_{od} - b\,t^g_{od}$ for the ground. She flies if and only if

$$\tau \leq (1-\theta)\, t^g_{od} \quad \Longleftrightarrow \quad \theta \leq s, \qquad \text{where} \quad \theta = \frac{P_{od} - q_{od}}{b\, t^g_{od}} \tag{6}$$

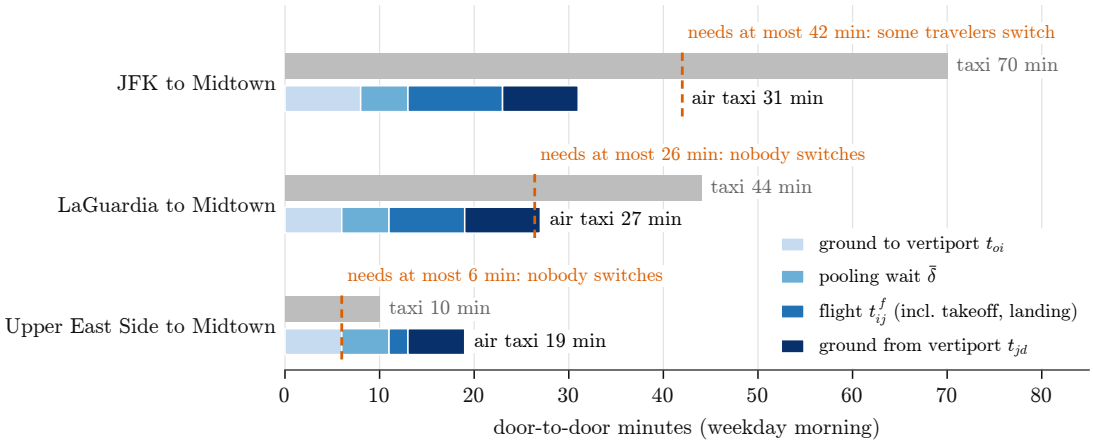


Figure 3: Air-taxi trip versus taxi in the NYC validation. Taxi times are observed. Air-taxi components are illustrative. The dashed line is the $\mu = 0.40$ threshold. Only long trips still save time after the fixed components.

is the *required relative time saving*. It is the extra price of flying, converted to minutes and divided by the ground time. With times in minutes, b is in dollars per minute. θ varies across travelers and trips, with mean μ and standard deviation σ. The share of travelers who fly is $\Pr(\theta \le s)$. We use two versions of this share. The first is the paper's distributionally robust (DRO) share. The second is an exact share under stated distributions. The exact share gives the main results.

DRO share (Proposition 1 of the paper). Suppose only μ and σ are known. The worst case over all distributions with these moments follows from Cantelli's one-sided Chebyshev inequality:

$$\inf_{F:\ \mathbb{E}_F\theta=\mu,\ \mathrm{Var}_F\theta=\sigma^2} \Pr_F(\theta \le s) = \begin{cases} \dfrac{(s-\mu)^2}{\sigma^2 + (s-\mu)^2} & s \ge \mu, \\ 0 & s < \mu. \end{cases} \tag{7}$$

With $\tilde{\tau} = (s-\mu)/\sigma$, the paper's share is

$$a^t_{oijd} = \frac{\tilde{\tau}^2}{1+\tilde{\tau}^2} \ \text{ if } \tilde{\tau} > 0, \qquad a^t_{oijd} = 0 \text{ otherwise}, \qquad \tilde{\tau} = \frac{s^t_{oijd} - \mu_{od}}{\sigma_{od}}. \tag{8}$$

For NYC, $\mu_{od} = \mu$ and $\sigma_{od} = \sigma$ on all pairs. For Chicago, both are estimated from data (Section 5). The bound is zero whenever $s \le \mu$. On most Chicago pairs $\mu_{od} > 1$, so the bound predicts almost no demand. For this reason, the main results use the exact share. Both shares are computed before the optimization, so the model stays linear. In the paper, τ depends on pooling and detour decisions. The S-curve is therefore part of the model and requires adaptive discretization.

Exact share (main results). Let $X = (P_{od} - q)/t$ be the extra price per minute. Then $\theta = X/b$, and a traveler chooses an option with saving s if and only if $X \le s\,b$. We assume X is normal on each OD pair, with moments from Section 5. We assume b is lognormal (median b_0, $\mathrm{CV}_b = 0.84$) and independent of X. Then

$$a^t_{oijd} = \mathbb{E}_b\left[\Phi\left(\frac{s^t_{oijd}\,b - \mathbb{E}[X]}{\mathrm{sd}(X)}\right)\right], \tag{9}$$

computed with 24-point Gauss–Hermite quadrature over $\ln b$. An option is offered only if $s^t_{oijd} \ge s_{\min} = 0.15$. This excludes travelers who would switch only for price. This happens when surge pricing makes the ground fare higher than the air-taxi fare.

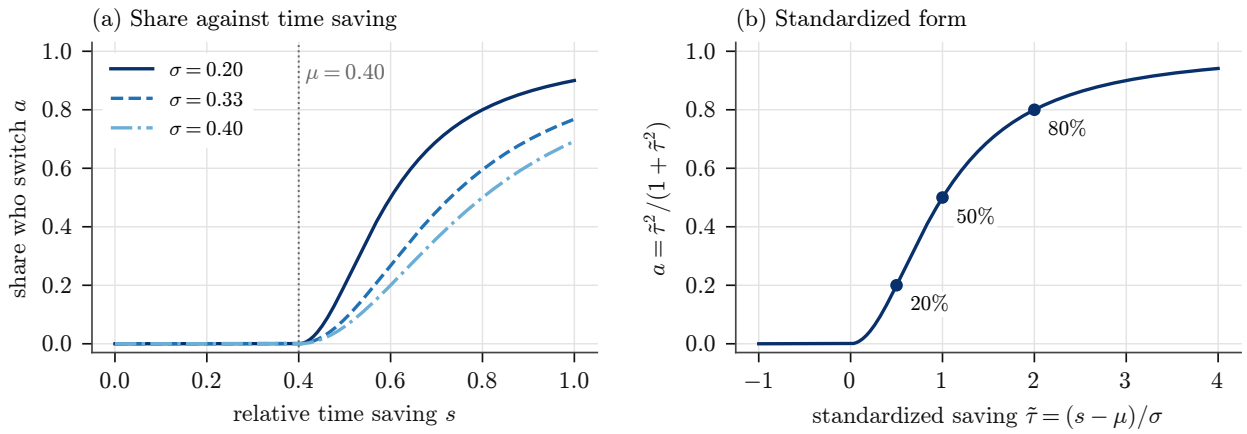


Figure 4: The DRO share (8). (a) Nobody switches until the saving exceeds μ. A larger σ makes the curve less steep. (b) In standardized units the curve is first convex, then concave. 20%, 50% and 80% of travelers switch at $\tilde{\tau} = 0.5$, 1 and 2.

4.5 Fares and margins

The fare is door to door and depends on the OD distance. The operator pays for the flight and both ground legs:

$$P_{od} = P_0 + r\,\ell_{od}, \tag{10}$$

$$c_{oijd} = \kappa\,\ell_{ij} + c^g\,(t^t_{oi} + t^t_{jd}) \qquad \text{(flight cost + ground-leg cost)}, \tag{11}$$

$$\pi_{oijd} = P_{od} - c_{oijd} \qquad \text{(margin per air-taxi passenger)}. \tag{12}$$

With $\pi_{oijd} = 1$ for all options, the objective counts passengers. All reported runs (NYC and Chicago) use this objective. The cost parameters κ, c^g and g_i are therefore not calibrated.

5 Customer expectations from trip data (RQ1)

For trip k on (o, d) in period t, with price q_k and ground time t_k:

$$\theta_k = \frac{P_{od} - q_k}{b\,t_k}. \tag{13}$$

Sufficient statistics. Downloading every trip would be slow. Instead, the City's data server returns five averages per (o, d, t):

$$m_1 = \mathbb{E}\left[\tfrac{1}{t}\right], \quad m_2 = \mathbb{E}\left[\tfrac{q}{t}\right], \quad m_3 = \mathbb{E}\left[\tfrac{1}{t^2}\right], \quad m_4 = \mathbb{E}\left[\tfrac{q}{t^2}\right], \quad m_5 = \mathbb{E}\left[\tfrac{q^2}{t^2}\right]. \tag{14}$$

For the extra price $X = (P_{od} - q)/t$, the first two moments follow exactly for *any* fare:

$$\mathbb{E}[X] = P_{od}\,m_1 - m_2, \qquad \mathbb{E}[X^2] = P_{od}^2\,m_3 - 2P_{od}\,m_4 + m_5. \tag{15}$$

Assume b is lognormal, with median b_0 and $s_b^2 = \ln(1 + \mathrm{CV}_b^2)$, and independent of the trip. Then $\mathbb{E}[1/b] = e^{s_b^2/2}/b_0$ and $\mathbb{E}[1/b^2] = e^{2s_b^2}/b_0^2$, so

$$\mu_{od} = \mathbb{E}[\theta] = \mathbb{E}[1/b]\,\mathbb{E}[X], \tag{16}$$

$$\sigma_{od}^2 = \mathbb{E}[1/b^2]\,\mathbb{E}[X^2] - \mu_{od}^2. \tag{17}$$

Fare rounding. Fares are rounded to the nearest \$2.50. Assume the rounding error ε is uniform on $[-1.25, 1.25]$ and independent of t. It then adds $\mathrm{Var}(\varepsilon)\,m_3$ to $\mathrm{Var}(X)$. The corrected values are

$$\mathrm{Var}^{\mathrm{corr}}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 - \frac{2.5^2}{12}\,m_3, \qquad \sigma_{od}^{2,\mathrm{corr}} = \sigma_{od}^2 - \frac{2.5^2}{12}\,\mathbb{E}[1/b^2]\,m_3. \tag{18}$$

Thin OD pairs. Some OD pairs have few trips (n_{od} in the month). For them, $\mathrm{sd}(X)$ is moved toward the period median m_t: $\widehat{\mathrm{sd}}(X) = \big(n_{od}\,\mathrm{sd}(X) + 30\,m_t\big)/(n_{od} + 30)$. The minimum is \$0.01 per minute. This is a simple shrinkage rule, in the spirit of Efron and Morris (1975). The weight 30 is our choice.

Uniform benchmark (the "average customer"). The benchmark keeps the same travelers but removes the OD-specific information. For the exact share, every OD pair receives the trip-weighted mixture of all OD distributions:

$$a^{\mathrm{uni}}(s) = \frac{\sum_{od} D_{od}\, a_{od}^{\mathrm{exact}}(s)}{\sum_{od} D_{od}}. \tag{19}$$

For the DRO share, the pooled mean is $\bar{\mu} = \sum_{od} D_{od}\mu_{od} / \sum_{od} D_{od}$. By the law of total variance, the pooled variance is $\bar{\sigma}^2 = \sum_{od} D_{od}(\sigma_{od}^2 + \mu_{od}^2) / \sum_{od} D_{od} - \bar{\mu}^2$. Both models then have the same average level. They differ only in heterogeneity.

Scenarios. Realistic fares give $\theta > 1$ on many OD pairs. Results are therefore reported for the four scenarios of Table 1, not for one fare (Figure 5).

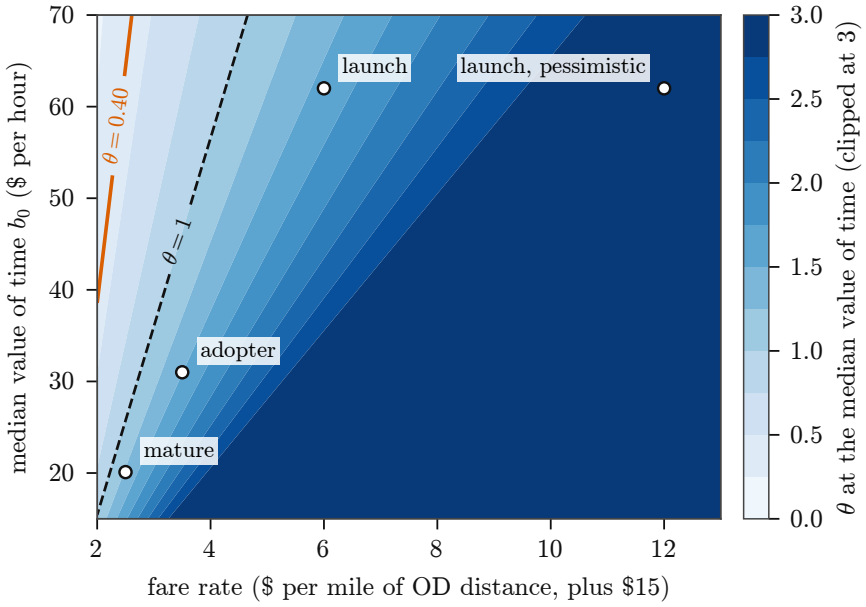


Figure 5: Required saving θ at the median value of time, over fare rates and values of time. The OD pair is illustrative: a 35-minute, \$30 rideshare trip of 12 miles. Dots show the four scenarios of Table 1. θ is close to the paper's 0.40 only with low fares and high values of time. Most Chicago switchers therefore have a value of time above the median.

6 Decision variables

$$y_i = \begin{cases} 1 & \text{if a vertiport is built in zone } i \\ 0 & \text{otherwise} \end{cases} \qquad i \in \mathcal{V}$$

x^t_{oijd} = share of the travelers from o to d in period t who are assigned to the vertiport pair $i \to j$ $\qquad (o,d) \in \mathcal{P},\ (i,j) \in \mathcal{K}^t_{od},\ t \in \mathcal{T}$

Option (o,i,j,d) carries $D^t_{od}\, a^t_{oijd}\, x^t_{oijd}$ air-taxi passengers per day in period t.

7 Optimization models

7.1 Model P0: NYC validation (Table 4 of the paper)

The objective is passengers minus a cost per vertiport:

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} D_{od}\, a_{oijd}\, x_{oijd} \;-\; c \sum_{i\in\mathcal{V}} y_i \tag{20}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d) \in \mathcal{P} \tag{21}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d) \in \mathcal{P},\ i \in \mathcal{V} \tag{22}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d) \in \mathcal{P},\ j \in \mathcal{V} \tag{23}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d) \tag{24}$$

Here c is a cost per vertiport in passenger units. It is calibrated so that $\mu = 0.40$ gives the paper's three vertiports. μ is then reduced from 0.40 to 0.33 with c fixed (Appendix A).

7.2 Model P1: one period (Chicago)

The objective is profit, or passengers with $\pi = 1$ and $g = 0$:

$$\max \quad \sum_{(o,d)\in\mathcal{P}} \sum_{(i,j)\in\mathcal{K}_{od}} \pi_{oijd}\, D_{od}\, a_{oijd}\, x_{oijd} \;-\; \sum_{i\in\mathcal{V}} g_i\, y_i \tag{25}$$

$$\text{s.t.} \quad \sum_{(i,j)\in\mathcal{K}_{od}} x_{oijd} \le 1 \qquad \forall (o,d) \in \mathcal{P} \tag{26}$$

$$\sum_{j:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_i \qquad \forall (o,d) \in \mathcal{P},\ i \in \mathcal{V} \tag{27}$$

$$\sum_{i:(i,j)\in\mathcal{K}_{od}} x_{oijd} \le y_j \qquad \forall (o,d) \in \mathcal{P},\ j \in \mathcal{V} \tag{28}$$

$$\sum_{i\in\mathcal{V}} y_i \le p \tag{29}$$

$$\frac{1}{H} \sum_{(o,d)\in\mathcal{P}} \sum_{(i',j')\in\mathcal{K}_{od}:\ i\in\{i',j'\}} D_{od}\, a_{oi'j'd}\, x_{oi'j'd} \;\le\; Q\, y_i \qquad \forall i \in \mathcal{V} \tag{30}$$

$$y_i \in \{0,1\}, \quad 0 \le x_{oijd} \le 1 \qquad \forall\, i,\ (o,i,j,d) \tag{31}$$

Constraint	Meaning
(26) Assignment	the shares of an OD pair sum to at most 1; the rest stays on the ground
(27) Departure link	travelers depart only from a built vertiport
(28) Arrival link	travelers arrive only at a built vertiport
(29) Budget	at most p vertiports (all reported runs, $p = 1, \ldots, 10$)
(30) Capacity	departing plus arriving passengers per hour at site i are at most Q; H is the length of the period in hours (optional)

7.3 Model P2: one network for all periods (RQ2)

$$\max \quad \sum_{t \in \mathcal{T}} w_t \sum_{(o,d) \in \mathcal{P}} \sum_{(i,j) \in \mathcal{K}^t_{od}} \pi^t_{oijd}\, D^t_{od}\, a^t_{oijd}\, x^t_{oijd} \;-\; G \sum_{i \in \mathcal{V}} g_i\, y_i \tag{32}$$

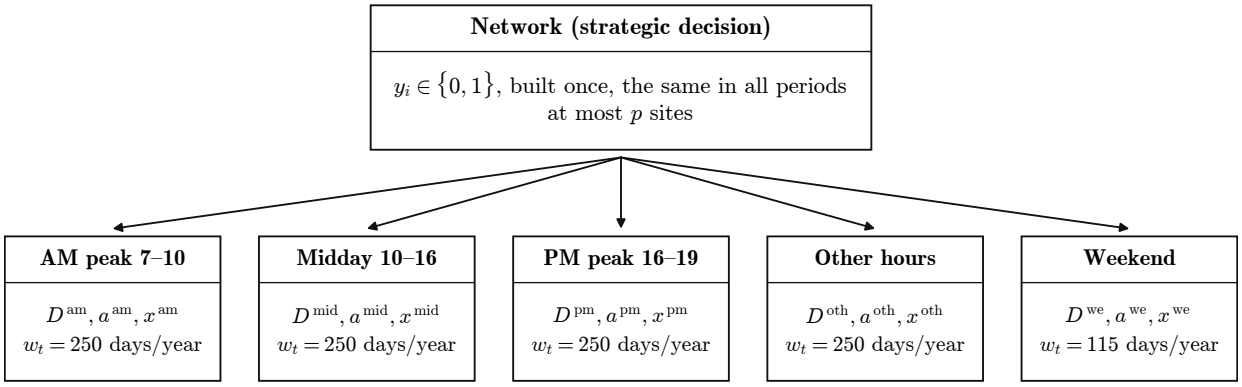
$$\text{s.t.} \quad \sum_{(i,j) \in \mathcal{K}^t_{od}} x^t_{oijd} \le 1 \qquad \forall (o,d),\ t \tag{33}$$

$$\sum_j x^t_{oijd} \le y_i, \qquad \sum_i x^t_{oijd} \le y_j \qquad \forall (o,d),\ i \text{ or } j,\ t \tag{34}$$

$$\text{capacity per period (optional)} \qquad \forall i,\ t \tag{35}$$

$$y_i \in \{0,1\}, \quad 0 \le x^t_{oijd} \le 1 \tag{36}$$

G is the number of operating days per year, so both terms are yearly. In the reported runs, $\pi = 1$ and $g = 0$, and a budget $\sum_i y_i \le p$ is added. The objective is then yearly passengers. The network y is the same in all periods. Only the routing x^t changes (Figure 6).



Robust plan: $y^R = \arg\max_y \sum_t w_t\, \Pi_t(y)$. Plan for one period t': $y^*_{t'} = \arg\max_y \Pi_{t'}(y)$.

Regret of planning for t': $\text{Reg}(t') = 1 - \sum_t w_t\, \Pi_t(y^*_{t'}) \,/\, \sum_t w_t\, \Pi_t(y^R)$.

Figure 6: Structure of Model P2 and comparison with single-period plans (passenger objective).

8 Model properties

Proposition 1 (The routing is integral for a fixed network)**.** *For any fixed $y \in \{0,1\}^{\mathcal{V}}$, Model P1 without capacity has an optimal solution in which each OD pair uses at most one vertiport pair, with $x_{oijd} \in \{0,1\}$. The optimal routing sends OD pair (o,d) through*

$$(i^*, j^*) \in \arg\max \{\pi_{oijd}\, a_{oijd} : (i,j) \in \mathcal{K}_{od},\ y_i = y_j = 1\}$$

if that maximum is positive, and keeps it on the ground otherwise.

Sketch. With y fixed, the problem separates by OD pair. If $y_i = 0$ or $y_j = 0$, the linking constraints force $x_{oijd} = 0$. Otherwise, the assignment constraint already implies them. The remaining set $\{x \geq 0 : \sum x \leq 1\}$ is a simplex. A linear objective reaches its maximum at a vertex of this set: all zeros or one option at 1. □

Remark. Proposition 1 gives the value of a network in closed form:

$$\Pi(y) = \sum_{(o,d)} D_{od} \max\Big\{0, \max_{(i,j)\in\mathcal{K}_{od}:\, y_i=y_j=1} \pi_{oijd}\, a_{oijd}\Big\} - \sum_i g_i y_i.$$

With this formula, any network can be evaluated quickly under any demand scenario. The metrics of Section 9 need this.

Proposition 2 (Strength of the linking constraints)**.**
The aggregated linking constraint $\sum_j x_{oijd} \leq y_i$ implies $x_{oijd} \leq y_i$ for every j. The reverse is not true. The formulation is therefore at least as tight as the classical "strong" facility-location formulation.

Structure and size. In P1, each customer (OD pair) needs *two* open facilities. This makes P1 closer to hub location (Campbell 1994; Alumur and Kara 2008) than to uncapacitated facility location (Cornuéjols, Nemhauser and Wolsey 1990). The main Chicago instance has 16 binaries and 2,289 options over all periods ($s \geq 0$). Of these, 617 weekday options have $s \geq 0.15$. The largest robustness instance has 77 binaries and about 36,000 options. Gurobi solves each instance in seconds. With 16 candidates and $p \leq 8$, there are 39,186 possible networks. Proposition 1 evaluates all of them exactly, and the results match the Gurobi solutions.

9 Evaluation metrics

Let $\Pi^S(y)$ be the value of network y under demand scenario S. The routing is optimal (Proposition 1).

Definition 1 (Value of heterogeneity, RQ1)**.** Let y^{het} be optimal with OD-specific shares. Let y^{unif} be optimal with the pooled shares of Section 5. We take the OD-specific model as the truth:

$$\text{VOH} = \Pi^{\text{het}}(y^{\text{het}}) - \Pi^{\text{het}}(y^{\text{unif}}) \;\geq 0, \qquad \text{VOH\%} = \frac{\text{VOH}}{\Pi^{\text{het}}(y^{\text{het}})}.$$

VOH is the loss from planning with one average expectation.

Definition 2 (Forecast error of the average-customer model, RQ1)**.** Take the network y^{unif}. FE is the relative gap between the pooled model's prediction and the OD-specific value: $\text{FE} = \big(\Pi^{\text{unif}}(y^{\text{unif}}) - \Pi^{\text{het}}(y^{\text{unif}})\big)/\Pi^{\text{het}}(y^{\text{unif}})$. A negative value means an under-forecast.

Definition 3 (Regret of a single-period plan, RQ2)**.** Let y^R solve P2. Let $y^*_{t'}$ solve P1 with the demand of period t' only. Both use the same budget p. The regret of planning for t' is the share of yearly passengers lost:

$$\text{Reg}(t') = 1 - \frac{\sum_t w_t\, \Pi_t(y^*_{t'})}{\sum_t w_t\, \Pi_t(y^R)} \;\geq 0.$$

With capacity, $\Pi_t(y^*_{t'})$ is computed by re-solving the routing with y fixed.

Network similarity. We compare two networks y and y' in two ways. The first is the Jaccard index $|S \cap S'|/|S \cup S'|$ of their site sets. The second is, for each site of y', the distance to the nearest site of y. We use both to compare with the paper and with Volakakis and Mahmassani (2024).

Benchmark: p-median (not run yet). This benchmark chooses p sites that minimize demand-weighted access distance:

$$\min \sum_o \sum_i D_o \, \ell_{oi} \, z_{oi} \quad \text{s.t. } \sum_i z_{oi} = 1,\ z_{oi} \le y_i,\ \sum_i y_i = p,\ y \in \{0,1\},\ z \ge 0,$$

Its network is then evaluated with $\Pi(y)$. The paper reports 22–53% lower profit for this benchmark.

10 Relation to the original model

Component	**Wang, Jacquillat and Vaze (2022)**	**This project**
Vertiport decisions	location and capacity level z_{ih}, fleet Γ	location y_i (capacity optional)
Matching	y_{oijd} binary, π_{oijd} fractions	x^t_{oijd} (integral at optimum)
Pooling wait	endogenous, $\delta_{ij} \ge (n-1)/\chi_{ij}$ (second-order cone)	fixed $\bar{\delta}$
Detours, reliability	endogenous (queueing network, M/M/1)	not modeled
Fleet, repositioning, charging	modeled	not modeled
Demand function	DRO or MNL S-curve inside the model	exact share (main) and DRO S-curve (comparison), precomputed; optional transfer penalty h
Customer expectations	uniform μ (0.40), σ	**uniform and OD-specific from fares (RQ1)**
Time of day	one demand snapshot (morning commute)	**five periods, robust network (RQ2)**
Solution method	adaptive discretization (MISOCP)	MILP, Gurobi

The simplified model is close to the paper's "exogenous operations" benchmark. A natural extension is to make the pooling wait endogenous again. Wang et al. (2022, Lemma 1) show how to write it as a second-order cone constraint.

11 Changes in version 1.1

Version 1 was written before the Chicago runs. Version 1.1 updates the document to match the code and the results:

- The exact share replaces the DRO share in the main results. Options need $s \ge s_{\min} = 0.15$ (Section 4.4).

- Expectations use a lognormal value of time and the scenarios of Table 1. Thin OD pairs are corrected, and pooled benchmarks are added (Section 5).
- All runs maximize passengers with a budget of p vertiports. Regret is a share of yearly passengers. A forecast-error metric is added (Section 9).
- The transfer penalty, capacity, bootstrap and seasonal check below are new.

11.1 Transfer penalty

The utility of Section 4.4 has no fixed cost for changing vehicles. The paper's equations (19)–(20) have none either. The paper's expectations come from surveys, which include such costs implicitly. Expectations from trip data do not include them. The penalty h is added to τ, with $h \in \{0, 5, 10, 15\}$ minutes and $s_{\min} \in \{0.15, 0\}$. Flight overhead and pooling wait also add fixed minutes to τ. Cruise speed has a similar effect. This analysis therefore covers them too.

11.2 Vertiport capacity

Constraint (30) is applied in P2 for every period. The period lengths are $H_t = 3$, 6, 3, 12 and 24 hours (AM, midday, PM, other weekday hours, weekend). Demand is distributed uniformly within each period. Capacity takes the values $Q \in \{\infty, 200, 100, 50\}$ passengers per hour (departing plus arriving). With capacity, the closed form of Proposition 1 no longer holds. Single-period networks are therefore evaluated by re-solving the routing with y fixed. The values of Q are illustrative.

11.3 Exact evaluation by enumeration and day-level bootstrap

Without capacity, Proposition 1 gives the value of every network in closed form. With 16 candidates and $p \leq 8$ there are 39,186 networks. All of them are evaluated directly. This reproduces the Gurobi solutions of P1 and P2 exactly. The 22 weekdays and 9 weekend days of March 2026 are resampled with replacement 200 times (Efron and Tibshirani 1993). Weekdays and weekend days are resampled separately. Days are resampled, not trips, because trips on the same day share traffic conditions. For each resample, demand, ground times, options and shares are recomputed. Every network is then evaluated again. Reported intervals are the 5th and 95th percentiles. They measure sampling variation in March-like conditions. They do not measure uncertainty in the assumed parameters.

11.4 Seasonal check

The same pipeline is applied to July 2025 and January 2026. For each p, the network planned for one month is evaluated on the other two. An all-season network maximizes the sum of the three monthly values. Each monthly value is divided by its own optimum, so all months have equal weight.

12 Runs reported

Run	Content	Model
V1	NYC validation: μ reduced from 0.40 to 0.33 with a calibrated cost c per vertiport (Appendix A)	P0
C1	RQ1: OD-specific vs pooled shares, four fare scenarios, five periods and weekday, $p = 1, \ldots, 10$; 16 sites and all 77 areas ($k = 3$ and $k = 8$)	P1
C2	RQ2: robust network vs single-period networks, three scenarios, $p = 2, \ldots, 8$; 16 sites and 77 areas	P2, P1
C3	Capacity $Q \in \{\infty, 200, 100, 50\}$, adopter scenario	P2
C4	Day-level bootstrap, 200 resamples, three scenarios	enumeration
C5	Transfer penalty $h \in \{0, 5, 10, 15\}$, $s_{\min} \in \{0.15, 0\}$	enumeration
C6	Seasonal check: July 2025, January 2026, March 2026	enumeration

Not yet run: p-median benchmark; spread of the value of time (CV_b from 0.5 to 1.2); March 2019 comparison; endogenous pooling.

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A NYC validation results (Model P0)

Yellow taxi, March 2019, weekdays 7–10 am, demand scaled by $S = 2$; 26 candidate zones; $v = 240$ km/h, $t_{\mathrm{oh}} = 10$, $\bar{\delta} = 5$, $\sigma = 0.20$; $c = 36.75$ passengers (any $c \in [32, 41.5]$ gives 3 vertiports at $\mu = 0.40$).

μ	**Sites (ours)**	**Sites (paper)**	**Passengers**	**Change in our network**
0.40	3	3	125.6	JFK, Times Sq/Theatre District, Gramercy
0.39	3	3	136.6	–
0.38	4	3	187.0	+ Upper East Side North
0.37	4	4	201.9	–
0.36	6	4	290.3	+ Kips Bay, LaGuardia, TriBeCa; − Gramercy
0.35	6	7	316.9	–
0.34	7	9	382.8	+ Gramercy, Midtown Center; − Kips Bay
0.33	7	10	416.7	–

The number of vertiports follows the paper closely. Most site differences are adjacent zones, for example Times Sq and Clinton East. The model puts the LaGuardia-area vertiport at the airport. The paper puts it in Astoria for operational reasons: shorter flights, higher reliability and shorter detours (Section 7.1.1 of the paper). This difference shows the effect of the operations layer, which this model does not include.